VISION Resistivity
Dual Frequency
Recorded Mode, Run 12

Schlumberger

Company:	Equinor
Well:	31/5-7
Field:	Eos
Rig Name:	West Hercules
Country:	Norway

Latitude:	60° 34' 35.14" N	UWID:	
Longitude:	3° 26' 36.13" E	Rig Name:	West Hercules
Block:	31/5	Rig Type:	Semi-Submersible

FL: Norwegian North Sea
FL1: Section: 8 1/2 in.
FL2: Job no.: 19SCA0085

Log Measured From: - Drill Floor: 31.00 m
Permanent Datum: - Mean Sea Level
Ground Level: 307.00 m

Acquisition Dates:	07-Jan-2020
Log Interval:	2629.00(m) -- 2711.00(m)
Index Types:	Measured Depth
Index Scales:	1:500
Depth Source:	Driller's Depth
Depth Sensor:	3rd Party Depth
Print Type:	Final
Spud Date:	02-Dec-2019

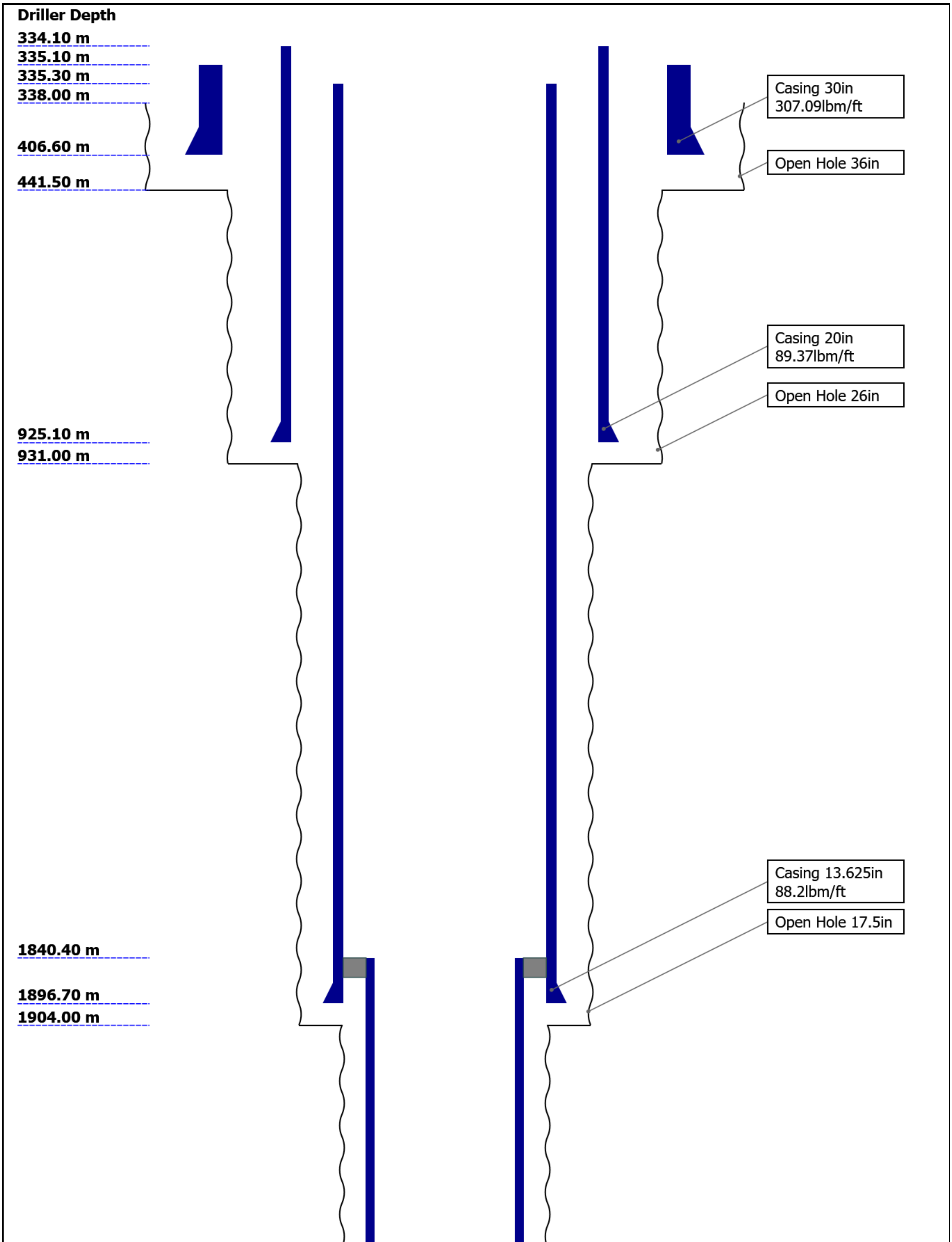
Other Services:
Direction & Inclination (Surveys)
Annular Pressure While Drilling
geoVISION Bit Resistivity

Disclaimer

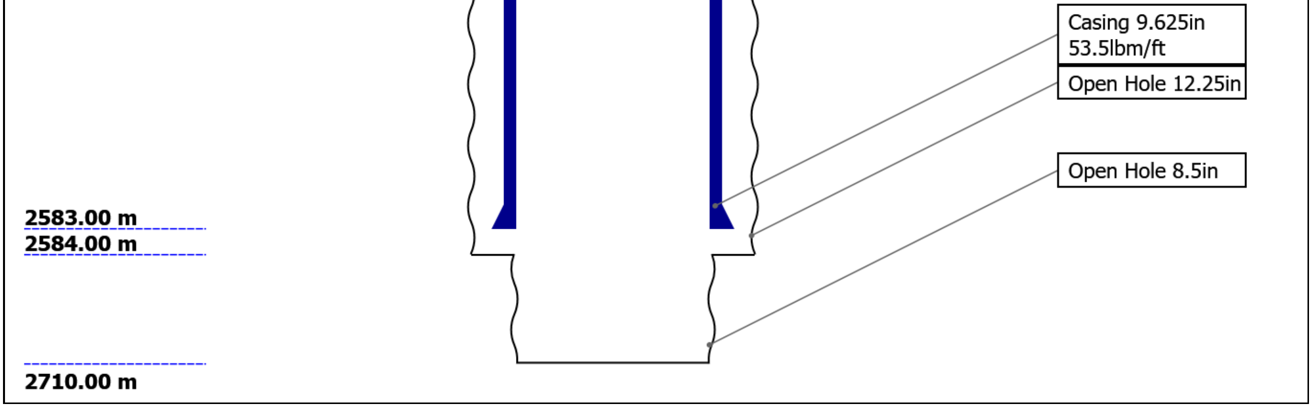
THE USE OF AND RELIANCE UPON THIS RECORDED-DATA BY THE HEREIN NAMED COMPANY (AND ANY OF ITS AFFILIATES, PARTNERS, REPRESENTATIVES, AGENTS, CONSULTANTS AND EMPLOYEES) IS SUBJECT TO THE TERMS AND CONDITIONS AGREED UPON BETWEEN SCHLUMBERGER AND THE COMPANY, INCLUDING: (a) RESTRICTIONS ON USE OF THE RECORDED-DATA; (b) DISCLAIMERS AND WAIVERS OF WARRANTIES AND REPRESENTATIONS REGARDING COMPANY'S USE AND RELIANCE UPON THE RECORDED-DATA; AND (c) CUSTOMER'S FULL AND SOLE RESPONSIBILITY FOR ANY INFERENCE DRAWN OR DECISION MADE IN CONNECTION WITH THE USE OF THIS RECORDED-DATA.

Contents

Well Sketch



Driller Depth
334.10 m
335.10 m
335.30 m
338.00 m
406.60 m
441.50 m
925.10 m
931.00 m
1840.40 m
1896.70 m
1904.00 m
Casing 30in
307.09lbm/ft
Open Hole 36in
Casing 20in
89.37lbm/ft
Open Hole 26in
Casing 13.625in
88.2lbm/ft
Open Hole 17.5in



Borehole Size/Casing Record

Bit						
Bit Size (in)	36	26	17.5	12.25	8.5	
Top Driller (m)	338	441.5	931	1904	2584	
Bottom Driller (m)	441.5	931	1904	2584	2710	
Casing						
Size (in)	30	20	13.625	9.625		
Weight (lbm/ft)	307.09	89.37	88.2	53.5		
Inner Diameter (in)	27.067	18.73	12.375	8.535		
Grade	X56	N80	P110	P110		
Top Driller (m)	335.1	334.1	335.3	1840.4		
Bottom Driller (m)	406.6	925.1	1896.7	2583		

Operational Run Summary

Parameter (unit)	**Run 12**					
Date Log Started	07-Jan-2020					
Time Log Started	04:05:33					
Date Log Finished	07-Jan-2020					
Time Log Finished	17:16:13					
Bit Size (in)	8.500					
Bit Start Depth (m)	2697.00					
Bit Stop Depth (m)	2710.00					
Top Log Interval (m)	2630.90					
Bottom Log Interval (m)	2709.03					
Max Hole Deviation (deg)	1.12					
Azimuth of Max Deviation (deg)	120.28					
Logging Unit Number	A615					
Logging Unit Location	Service Office					
Recorded By	M.Martter					
Witnessed By	R.Nordstokke					
Service Order Number	19SCA0085					

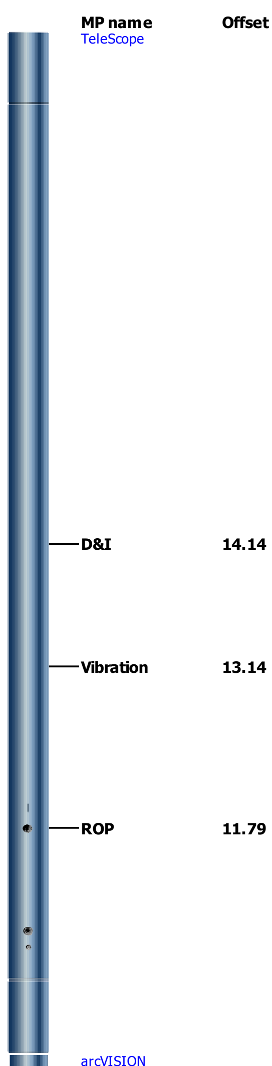
Borehole Fluids

Parameter(unit)	Run 12						
Fluid Type	Water						
Fluid Name	Glydril						
Max Recorded Temperatures (degC)	44						
Source of Sample	Active Tank						
Salinity (ppm)	97140						
Density (g/cm3)	1.15						
Funnel Viscosity (s)							
Fluid Loss (cm3)							
PH							
Source RMF							
RMC	Pressed						
RM @ Meas Temp (ohm.m@degC)	0.08 @ 14.8						
RMF @ Meas Temp (ohm.m@degC)	0.07 @ 14.8						
RMC @ Meas Temp (ohm.m@degC)	0.09 @ 20						
RM @ BHT (ohm.m@degC)	0.04 @ 44						
RMF @ BHT (ohm.m@degC)	0.04 @ 44						
RMC @ BHT (ohm.m@degC)	0.06 @ 44						
Total Solid (%)							
High Gravity Solids (%)							

Remarks and Equipment Summary

Run 12: Toolstring

Equip name	Length
TELE675:E4388	**18.37**
MSSU675:30848	
Upper Extender	
MDC675:E4388	
MMA:2507	
MDI:3666	
PMVC:236	
PMEA:803	
MTA:2162	
MTK675:004	
MSSD675:1005482-1	
Lower Extender, Long	
ARC6:H0143-1	**9.91**



Run 12: Remarks

All depths referenced to driller's depth. See EOWR for depth tracking details.

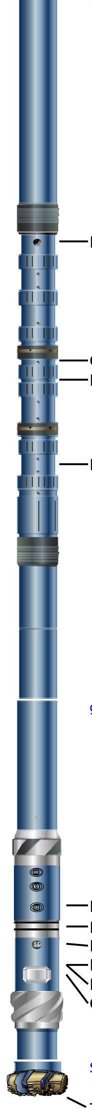
All data from tool memory. All data acquired while drilling.

Gamma Ray measurement is environmentally corrected for mud weight, bit size, collar thickness and Potassium content.

Resistivity measurements are borehole compensated and environmentally corrected for bit size, mud resistivity and temperature.

Data acquired while reaming down over cored interval from 2643 to 2697 m bit depth.

Run 12 POOH at 2710 m to perform coring.



Lengths are in m
Maximum Outer Diameter = 8.500 in
Line: Sensor Location, Value: Gating Offset
All measurements are relative to TOOL_ZERO

Run 12

VISION Resistivity - Dual Frequency

Software Version

Acquisition System	Version
Maxwell 2019.2	9.2.113335.3100
Application Patch	DnM_TestKit-XceedCPU41-2019.2_9.2.114181

Computation	Description	Version
ARCResistivity	ARC Resistivity Computation Package for ARC Tool Family	9.2.113335.3100
ARC6GammaRay	ARC6 Gamma Ray Computation Package for both Real-time and Recorded Mode	9.2.113335.3100
GVRRes	GVR Resistivity Computation Package for both Real-time and Recorded Mode	9.2.113335.3100

Tool Interface	System Version	Loaded Version
HSPM	2019.0c.001	9.2.113335.3100

Tool Elements	Description	Software Version	Firmware Version
RBEC	Electronics Chassis Assembly for RAB6-C	9.2.113335.3100	V8.8B
DRILLING_SURFACE	DRILLING_SURFACE	9.2.113335.3100	
ARDC	ARC 6.75 Inch Tool Drill Collar	9.2.113335.3100	V9.7B

Pass Summary

Run Name	Pass Objective	Direction	Top	Bottom	Start	Stop	Include Parallel Data
Run 12	Drilling	Down	298.98 m	2709.75 m	07-Jan-2020 4:05:33 AM	07-Jan-2020 5:16:13 PM	Yes

All depths are referenced to toolstring zero

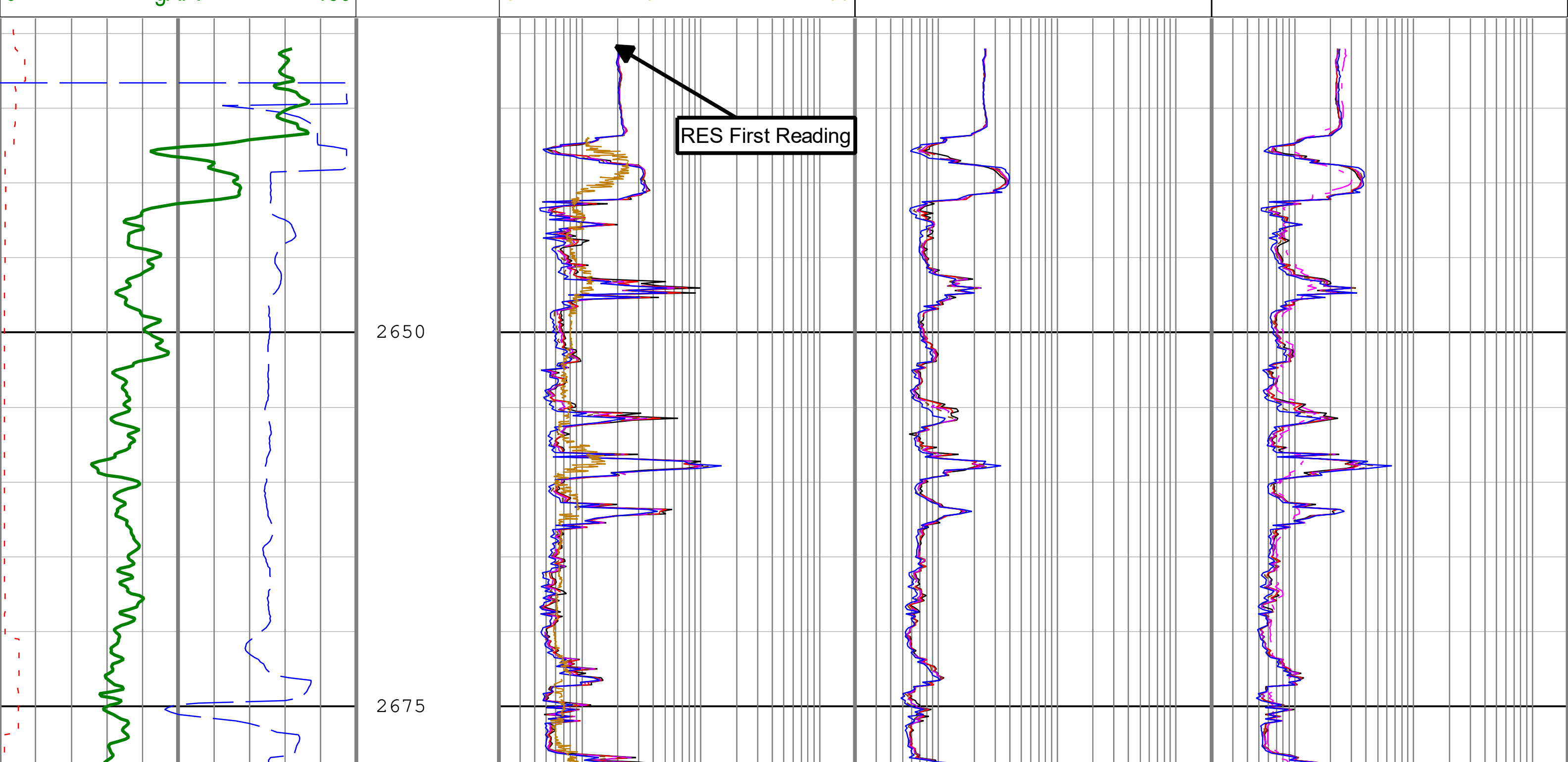
Log

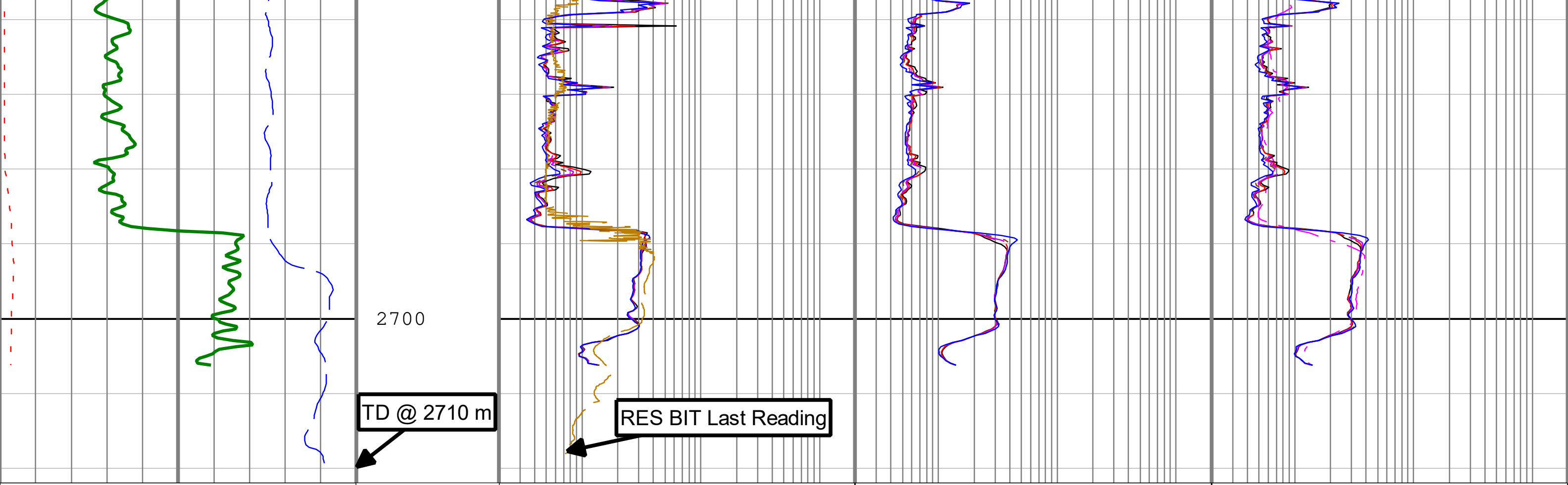
Company:Equinor Well:31/5-7

Run 12: Drilling:S021

Description: Format: Log (Resistivity - Dual Frequency) Index Scale: 1:500 Index Unit: m Index Type: Measured Depth Creation Date: 20-Jan-2020 13:03:26

Track	Curve	Scale
1	Resistivity Time After Bit (ARTM) ARC6	0 h 10
1	Rate of penetration averaged over the last 5 ft (1.5 m) (ROP5) RT	200 m/h 0
1	Gamma Ray (GR_ARC) ARC6 RM	0 gAPI 150
2	Phase Shift Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (P40H) ARC6 RM	0.2 ohm.m 200
2	Phase Shift Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected. (P34H) ARC6 RM	0.2 ohm.m 200
2	Phase Shift Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected. (P28H) ARC6 RM	0.2 ohm.m 200
2	Phase Shift Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected. (P22H) ARC6 RM	0.2 ohm.m 200
2	Phase Shift Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected. (P16H) ARC6 RM	0.2 ohm.m 200
2	Bit Resistivity (RES_BIT) RAB6 RM	0.2 ohm.m 200
3	Attenuation Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (A40H) ARC6 RM	0.2 ohm.m 200
3	Attenuation Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected (A34H) ARC6 RM	0.2 ohm.m 200
3	Attenuation Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected (A28H) ARC6 RM	0.2 ohm.m 200
3	Attenuation Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected (A22H) ARC6 RM	0.2 ohm.m 200
3	Attenuation Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected (A16H) ARC6 RM	0.2 ohm.m 200
4	Attenuation Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected (A40L) ARC6 RM	0.2 ohm.m 200
4	Phase Shift Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected. (P40L) ARC6 RM	0.2 ohm.m 200
4	Phase Shift Resistivity 34 inch Spacing at 400 KHz, Environmentally Corrected. (P34L) ARC6 RM	0.2 ohm.m 200
4	Phase Shift Resistivity 28 inch Spacing at 400 KHz, Environmentally Corrected. (P28L) ARC6 RM	0.2 ohm.m 200
4	Phase Shift Resistivity 22 inch Spacing at 400 KHz, Environmentally Corrected. (P22L) ARC6 RM	0.2 ohm.m 200
4	Phase Shift Resistivity 16 inch Spacing at 400 KHz, Environmentally Corrected. (P16L) ARC6 RM	0.2 ohm.m 200





Resistivity Time After Bit (ARTM) ARC6
0 h 10

Rate of penetration averaged over the last 5 ft (1.5 m) (ROP5) RT
200 m/h 0

Gamma Ray (GR_ARC) ARC6 RM
0 gAPI 150

Phase Shift Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (P40H) ARC6 RM
0.2 ohm.m 200

Phase Shift Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected. (P34H) ARC6 RM
0.2 ohm.m 200

Phase Shift Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected. (P28H) ARC6 RM
0.2 ohm.m 200

Phase Shift Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected. (P22H) ARC6 RM
0.2 ohm.m 200

Phase Shift Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected. (P16H) ARC6 RM
0.2 ohm.m 200

Bit Resistivity (RES_BIT) RAB6 RM
0.2 ohm.m 200

Attenuation Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (A40H) ARC6 RM
0.2 ohm.m 200

Attenuation Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected (A34H) ARC6 RM
0.2 ohm.m 200

Attenuation Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected (A28H) ARC6 RM
0.2 ohm.m 200

Attenuation Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected (A22H) ARC6 RM
0.2 ohm.m 200

Attenuation Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected (A16H) ARC6 RM
0.2 ohm.m 200

Attenuation Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected (A40L) ARC6 RM
0.2 ohm.m 200

Phase Shift Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected. (P40L) ARC6 RM
0.2 ohm.m 200

Phase Shift Resistivity 34 inch Spacing at 400 KHz, Environmentally Corrected. (P34L) ARC6 RM
0.2 ohm.m 200

Phase Shift Resistivity 28 inch Spacing at 400 KHz, Environmentally Corrected. (P28L) ARC6 RM
0.2 ohm.m 200

Phase Shift Resistivity 22 inch Spacing at 400 KHz, Environmentally Corrected. (P22L) ARC6 RM
0.2 ohm.m 200

Phase Shift Resistivity 16 inch Spacing at 400 KHz, Environmentally Corrected. (P16L) ARC6 RM
0.2 ohm.m 200

Description: Format: Log (Resistivity - Dual Frequency) Index Scale: 1:500 Index Unit: m Index Type: Measured Depth Creation Date: 20-Jan-2020 13:03:26

Channel Processing Parameters

Run 12: Parameters

Parameter	Description	Tool	Value	Unit
ABNT	Abnormal Transmitter Indicator	ARC6	NO_TX_FAILED	
BHK	Drilling Fluid Potassium Concentration	Borehole	6	%
BHT	Bottom Hole Temperature	Borehole	44	degC
BS	Bit Size	DNMSESSION	8.5	in
DFD	Drilling Fluid Density	Borehole	1.15	g/cm3
DFT_CATEGORY	Drilling Fluid Type	Borehole	Water	

GRSE_RM	Generalized Mud Resistivity Selection for Recorded Mode	Borehole	REMS(RM)	
GTSE_RM	Generalized Temperature Selection for Recorded Mode	Borehole	DHAT(RM)	
HIGH_BLEND	High Resistivity Threshold for Blending	ARC6	2	ohm.m
LOW_BLEND	Low Resistivity Threshold for Blending	ARC6	1	ohm.m
MST	Mud Sample Temperature	Borehole	14.8	degC
RMS	Resistivity of Mud Sample	Borehole	0.08	ohm.m
TEMP_SEL_RAB	RAB Temperature Selection	RAB6	Tool	

Tool Control Parameters

Calibration Report

RAB6 (GeoVision Resistivity 675) Calibration - Run 12

Primary Equipment :

Electronics Chassis	RBEC	216

M21V - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit
Monitor 2 at T1 Calibration Coefficient		Master	1.0000	0.9750	1.0000	1.0250

M22V - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit
Monitor 2 at T2 Calibration Coefficient		Master	1.0000	0.9750	1.0020	1.0250

M01V - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit
Monitor 0 at T1 Calibration Coefficient		Master	1.0000	0.9750	0.9950	1.0250

M02V - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit
Monitor 0 at T2 Calibration Coefficient		Master	1.0000	0.9750	0.9980	1.0250

R1V - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit
Ring at T1 Calibration Coefficient		Master	1.0000	0.9750	1.0000	1.0250

R2V - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit
Ring at T2 Calibration Coefficient		Master	1.0000	0.9750	1.0030	1.0250

BDM1 - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit
Button Deep at T1 Calibration Coefficient		Master	1.0000	0.9750	1.0070	1.0250

BDM2 - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit
Button Deep at T2 Calibration Coefficient		Master	1.0000	0.9750	1.0100	1.0250

BMM1 - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit
Button Medium at T1 Calibration Coefficient		Master	1.0000	0.9750	0.9990	1.0250

BMM2 - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit
Button Medium at T2 Calibration Coefficient		Master	1.0000	0.9750	1.0020	1.0250

BSM1 - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit
Button Shallow at T1 Calibration Coefficient		Master	1.0000	0.9750	1.0000	1.0250

BSM2 - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit
Button Shallow at T2 Calibration Coefficient		Master	1.0000	0.9750	1.0030	1.0250

PGR - Gamma Ray: Blanket

Master (Time Frame File): 14:06:21 18-Dec-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit
Gamma Ray Calibration Gain		Master	1.0000	0.7500	0.9838	1.2500

ARC6 (Array Resistivity Compensated 675) Calibration - Run 12

Primary Equipment :

Elec. Chassis HP w/o AIM Receiver AREA 618

RESAIRCAL - Resistivity: Air

Master (Time Frame File): 15:26:50 07-Dec-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit
Attenuation T1 at 2 MHz	dB	Master	8.500	6.500	7.951	10.500
Attenuation T2 at 2 MHz	dB	Master	6.500	4.500	6.997	8.500
Attenuation T3 at 2 MHz	dB	Master	4.500	2.500	4.584	6.500
Attenuation T4 at 2 MHz	dB	Master	4.600	2.600	4.904	6.600
Attenuation T5 at 2 MHz	dB	Master	3.600	1.600	3.142	5.600
Phase Shift T1 at 2 MHz	deg	Master	0.100	-3.900	1.927	4.100
Phase Shift T2 at 2 MHz	deg	Master	0.100	-3.900	-1.871	4.100
Phase Shift T3 at 2 MHz	deg	Master	0.100	-3.900	1.878	4.100
Phase Shift T4 at 2 MHz	deg	Master	0.100	-3.900	-1.879	4.100
Phase Shift T5 at 2 MHz	deg	Master	0.100	-3.900	1.856	4.100
Attenuation T1 at 400 KHz	dB	Master	8.500	6.500	7.970	10.500
Attenuation T2 at 400 KHz	dB	Master	6.500	4.500	6.989	8.500
Attenuation T3 at 400 KHz	dB	Master	4.500	2.500	4.593	6.500
Attenuation T4 at 400 KHz	dB	Master	4.600	2.600	4.889	6.600
Attenuation T5 at 400 KHz	dB	Master	3.600	1.600	3.160	5.600
Phase Shift T1 at 400 KHz	deg	Master	0.100	-3.900	-0.666	4.100
Phase Shift T2 at 400 KHz	deg	Master	0.100	-3.900	0.584	4.100
Phase Shift T3 at 400 KHz	deg	Master	0.100	-3.900	-0.629	4.100
Phase Shift T4 at 400 KHz	deg	Master	0.100	-3.900	0.582	4.100
Phase Shift T5 at 400 KHz	deg	Master	0.100	-3.900	-0.643	4.100

GRGAIN - Gamma Ray: Blanket

Master (Time Frame File): 10:23:30 06-Jun-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit
Gamma Ray Calibration Gain		Master	1.000	0.580	1.100	1.250

Company:	Equinor
Well:	31/5-7

Field:	Eos
Rig Name:	West Hercules
Country:	Norway

Schlumberger

VISION Resistivity

Dual Frequency

Recorded Mode, Run 12